

Contract Adjustment and Joint Output Responses in Vertical Oligopoly

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Abstract

When producers share suppliers, similar output responses to a cost shock can arise under different input-pricing mechanisms. We derive joint-response restrictions that competitive supply must satisfy and show that bargaining can violate them even when individual response signs coincide. Starting from the same equilibrium, we compare firms' output responses to the same cost shocks with all input prices fixed and with those prices allowed to adjust. From these responses, we recover feedback through input costs: under competitive supply, it must be nonnegative and equal in both directions between buyers after weighting by their input expenditures. The result holds under common marginal-cost pricing, separable convex supplier costs, substitutable CES inputs, fixed outside quantities, and regular local equilibria. Numerically validated bargaining examples violate the sign or weighted-reciprocity restriction. An application disciplined by high-bandwidth memory supply quantities shows why individual signs can mislead: contract adjustment can attenuate rival expansion under either supply mechanism, while bargaining violates reciprocity across the customer allocations examined. The application illustrates the restrictions and their quantitative implications; empirical use requires responses under known contract states.

Keywords: vertical bargaining, joint output responses, contract adjustment, competitive supply, semiconductors.

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1 Introduction

When producers share suppliers, a shock to one producer affects its rivals through both product-market competition and input prices. We derive restrictions on joint output responses under fixed and reset input prices that competitive supply must satisfy, and show how bargaining can violate them. Comparing the two contract states at the same baseline isolates restrictions on the supply mechanism while allowing downstream strategic interaction. These restrictions can distinguish mechanisms that generate the same individual response signs, including cases in which contract adjustment attenuates a rival’s expansion.

Vertical bargaining research explains how disagreement behavior and trading alternatives shape negotiated prices and transmit shocks across buyers (Iozzi and Valletti, 2014; Inderst and Valletti, 2011; Alviarez et al., 2026). Our contribution connects these mechanisms to restrictions on the full set of buyer-specific output responses across contract states. A common marginal-cost supplier must generate reciprocal cost feedback, after weighting by buyer input expenditure, and nonnegative feedback under the maintained input-substitution structure. Negotiated terms can break these restrictions. This gives economic content to comparing the two directions of response: a rival’s expansion or cost relief alone can be compatible with both supply mechanisms, while their joint responses can reject the competitive benchmark.

The diagnostic builds on cost duality and substitution, including the connection between substitutes and inverse monotonicity (Galichon et al., 2025). Its contribution is to express the resulting supply restrictions through output responses under two contract states. Related conduct analysis uses differences in inverse price pass-through matrices (Dearing et al., 2026). Here the inverse fixed-price and reset-price output responses share one baseline and one downstream response, which cancels in their difference. The result specifies necessary conditions for the maintained competitive-supply structure and the response information needed to assess them.

The construction applies a separate, small cost wedge to each buyer. Let H collect output responses when all input unit prices remain at their baseline values, and let B collect responses when all active prices reset. With common supplier marginal-cost pricing, separable convex costs, substitutable CES inputs, fixed outside quantities, and nonsingular local responses, $N = H^{-1} - B^{-1}$ is entrywise nonnegative. Weighting its rows by input expenditure makes it symmetric and positive semidefinite. Small bargaining examples show that individual own and cross responses can retain their signs while N violates nonnegativity or weighted reciprocity. These examples establish that the competitive restrictions need not survive bargaining; they do not impose a general sign on bargaining responses.

Downstream conduct is reflected separately in H . Cournot competition generally gives nonzero cross responses, whereas the independent-demand constant-markup comparator makes H diagonal. Both demand systems can transmit shocks through negotiated input prices once contracts reset. Thus the supply restriction and the downstream comparison concern different features of the model. Empirical implementation would require buyer-specific shock responses and matched input expenditures under known fixed-price and reset-price states. Selective contract renewal changes this comparison and is outside the full-reset restriction.

High-bandwidth memory (HBM) provides an application in which qualified sourcing links and advance contracting make these distinctions economically relevant. Product-level supply information disciplines physical quantity targets, while customer allocations, elasticities, and bargaining weights remain imposed. In the calibrated examples, individual response signs do not separate the supply mechanisms: price adjustment can attenuate rival expansion under both common marginal-cost pricing and bargaining. Across three customer allocations, however, the bargaining examples violate weighted reciprocity while the competitive examples satisfy it to numerical precision. The application demonstrates the diagnostic in economies disciplined by supply quantities; those quantities do not themselves identify H or B .

The application also clarifies quantitative incidence. At the dense bargaining–Cournot baseline, a five-percent targeted cost wedge makes price adjustment add 3.06 percentage points to rival output. Removing the target’s Samsung link at fixed primitives reduces that addition to 1.29 points. Refitting the sparse economy to the original quantity targets instead gives -1.48 points. Rival output rises in all three comparisons. Customer-allocation sensitivity and a cost–productivity decomposition show why the change in the additional effect cannot be attributed to the missing link alone. This distinction matters for policy analysis with endogenous input prices (Spencer and Jones, 1992; Weyl and Fabinger, 2013; Grossman et al., 2024): a network counterfactual at fixed primitives and a comparison of separately fitted economies answer different questions. The application specifies which sourcing and contract information would be needed to turn these conditional calculations into an industry assessment.

Sections 2–3 develop the model and joint-response restrictions. Sections 4–5 present the HBM application; the appendices provide proofs, numerical validation, and supplementary outcomes.

2 A common vertical model and two contract states

There are two downstream buyers, $d \in \{t, r\}$, and a set of upstream suppliers u . An active link has a fixed weight $\alpha_{du} > 0$; an unavailable link has weight zero. The set of available suppliers and their potential weights remain fixed within a policy experiment. Removing a link does not renormalize the weights of the remaining inputs. Productivities and demand scales satisfy $z_d, A, B_d > 0$, and the demand elasticities satisfy $\sigma, \rho, \nu > 1$; the Cournot comparisons below use $\rho \leq \sigma$.

2.1 Technology, quantities, and costs

The materials price index and the physical input demand associated with output q_d are

$$P_d^M = \left(\sum_u \alpha_{du} p_{du}^{1-\eta} \right)^{1/(1-\eta)}, \quad (1)$$

$$x_{du} = \frac{q_d}{z_d} \alpha_{du} \left(\frac{p_{du}}{P_d^M} \right)^{-\eta}, \quad \eta > 1. \quad (2)$$

Consequently $\sum_u p_{du} x_{du} = P_d^M q_d / z_d$. Write $\mu_d = P_d^M / z_d$ for expenditure per physical unit of final output and $E_d = \mu_d q_d$ for materials expenditure. Thus P_d^M prices a unit of the input composite, whereas μ_d incorporates the physical productivity conversion. The buyer's effective unit cost is

$$C_d = \frac{\tau_d}{z_d} P_d^M. \quad (3)$$

The friction τ_d increases the effective cost of producing final output. It does not multiply physical HBM requirements in (2) and is therefore not a physical iceberg loss. Payments to suppliers remain $p_{du} x_{du}$. Any additional burden represented by τ_d is outside those receipts; we do not assign it a tax recipient or use it to calculate social welfare.

Focal and total supplier volumes are $Q_u^F = \sum_d x_{du}$ and $Q_u^T = Q_u^F + Q_u^O$. Outside-customer volume Q_u^O is fixed. With supplier scale $c_u > 0$ and slope $\psi_u \geq 0$, the cost attributable to serving focal buyers is

$$K_u^F = c_u Q_u^F + c_u \psi_u \left(Q_u^O Q_u^F + \frac{1}{2} (Q_u^F)^2 \right), \quad MC_u = c_u (1 + \psi_u Q_u^T). \quad (4)$$

This incremental cost includes the interaction with outside production. The scale c_u is a cost per unit of supplier output, distinct from the buyer's unit materials cost μ_d . The core exercise uses a common floor c_u across each supplier's buyers. Proposition 1 maintains this common-price restriction; buyer-specific floors fall outside its assumptions.

2.2 Downstream competition

In the Cournot specifications, inverse demands are

$$p_d^D = A q_d^{-1/\sigma} Q^{1/\sigma-1/\rho}, \quad Q = \left(\sum_j q_j^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (5)$$

Define $s_d = q_d^{(\sigma-1)/\sigma} / \sum_j q_j^{(\sigma-1)/\sigma}$. Interior quantity choices satisfy

$$p_d^D \left[1 - \frac{1}{\sigma} - \left(\frac{1}{\rho} - \frac{1}{\sigma} \right) s_d \right] = C_d. \quad (6)$$

Here s_d also equals the model's revenue share; an empirical revenue share measures this object only if its product and market scope agrees. A rival's share response and quantity response are distinct because aggregate demand also changes.

The constant-markup comparator instead uses independent demands $q_d = B_d (p_d^D / A)^{-\nu}$ and prices $p_d^D = \nu C_d / (\nu - 1)$. For $B_d = 1$ and $\sigma = \rho = \nu$, (5) has this same independent-demand limit. This is a demand-block comparison, not a claim that the entire framework nests another published bargaining model.

2.3 Prices and disagreement

Downstream profit is $\Pi_d^D = (p_d^D - C_d)q_d$. Focal upstream profit is $\Pi_u^F = \sum_d p_{du}x_{du} - K_u^F$. Under bargaining, each active pair chooses a linear price to maximize

$$(G_{du}^D)^\gamma (G_{du}^U)^{1-\gamma}, \quad 0 < \gamma < 1, \quad (7)$$

where a gain is the firm's agreement profit less its profit when the link is removed. The other contract prices remain fixed in that disagreement calculation, but all downstream quantities and physical input demands are recomputed. A buyer with no remaining input supplier exits. Outside-customer quantities, the surviving α weights, and cost scales are not changed after disagreement. There are no additional lump-sum or cross-link transfers.

The bargaining equilibrium jointly solves all active price equations and the downstream choices. We require positive bilateral gains and check the local Nash conditions and a finite search over unilateral price deviations, from one hundredth to one hundred times each candidate price. The numerical exercise follows interior candidates satisfying these checks; the finite search does not certify a global best response, and joint equilibrium uniqueness is not established. The Nash-in-Nash foundation in [Collard-Wexler et al. \(2019\)](#) motivates the solution concept, but is not invoked as a proof that every condition of that paper holds in this quantity and linear-price environment. Disagreement behavior is substantive, as emphasized by [Iozzi and Valletti \(2014\)](#).

The marginal-cost benchmark replaces (7) by $p_{du} = MC_u$ on active links, with the same physical demand and cost curves. Positive supply slopes can transmit a buyer shock to other buyers even without a bargaining margin. We refer to this as the marginal-cost-pricing benchmark: it is a conduct restriction applied to a fixed supplier network, not a model of free entry into supplier relationships.

2.4 The policy experiment

All baselines have $\tau_d = 1$. For the joint-response comparison, each buyer's wedge is perturbed in turn at that same baseline. A finite policy experiment raises only the targeted buyer's τ_t , holding fitted and imposed primitives fixed. Under *fixed unit prices*, baseline link prices are retained and quantities reoptimize. This state imposes neither a new Nash condition nor post-shock marginal-cost pricing. If inherited prices generate losses, they are reported rather than silently reset. Under *reset prices*, all active prices satisfy the selected conduct rule at the new quantities.

Both experiments condition on the existing qualification set and fixed outside volumes. They omit inventory adjustment, capacity rationing, enforceable fixed allocations, and new qualification investment. Thus they describe two contract-adjustment comparisons within a sourcing structure, not the full short-run and long-run equilibria of the industry.

3 Joint output responses and competitive-supply restrictions

A rival can expand under both competitive supply and bargaining. The joint responses to shocks to each buyer contain more information: comparing fixed and reset prices removes the common downstream response and exposes restrictions on the input-supply mechanism.

3.1 Responses at a common baseline

Fix one model's interior baseline with $\tau = \mathbf{1}$. Let $q^F(\tau)$ retain all baseline input prices and let $q^R(\tau)$ reset all active prices under that model's conduct rule. Define

$$H_{ij} = \left. \frac{\partial \log q_i^F}{\partial \log \tau_j} \right|_1, \quad B_{ij} = \left. \frac{\partial \log q_i^R}{\partial \log \tau_j} \right|_1. \quad (8)$$

Every column is a separate buyer's shock. Technology, demand, qualification weights, and outside quantities are fixed across the two contract states. Since the wedge enters only $C_i = \tau_i \mu_i$, the fixed-price matrix also equals $H_{ij} = \partial \log q_i / \partial \log C_j$. Both matrices use the same baseline prices and quantities within each model; matrices from separately calibrated models are not combined.

When H and B are nonsingular, form

$$N = H^{-1} - B^{-1}, \quad S = \text{diag}(E)N, \quad E_i = \mu_i q_i = \sum_u p_{iu} x_{iu}. \quad (9)$$

The weights are materials expenditures for the input scope in the model, rather than sales or total accounting costs. N is dimensionless; S has expenditure units.

Proposition 1 (Competitive-supply restrictions on joint responses). *Maintain the fixed common- η CES technology in (1)–(2), fixed outside quantities, and separable twice-differentiable supplier costs $G_u(Q_u^T)$ with $G'_u > 0$ and $G''_u \geq 0$. All customers of a supplier pay its common marginal cost $p_{iu} = p_u = G'_u(Q_u^T)$. Suppose the baseline and nearby equilibria are interior and differentiable, and H and B are nonsingular. Then*

$$N_{ij} \geq 0 \quad \text{for every } i, j, \quad S = S', \quad S \succeq 0. \quad (10)$$

In particular, $E_i N_{ij} = E_j N_{ji}$. The quadratic cost specification (4) satisfies the maintained supply conditions.

Economic argument. At a given output vector, competitive sourcing minimizes a common convex resource-cost function. Its gradient is the vector of unit materials costs μ , so $K = \partial \mu / \partial q$ is symmetric and positive semidefinite. Conditional CES substitution and nonnegative supply slopes also imply $K_{ij} \geq 0$; Appendix A derives this property directly from input demands and marginal-cost pricing. Write $y = d \log q$, $v = d \log \tau$, and $e = d \log \mu$. The downstream equation is $y = H(v + e)$, while reset supply gives $e = \text{diag}(\mu)^{-1} K \text{diag}(q) y$. Since $y = Bv$, eliminating v yields

$$N = \text{diag}(\mu)^{-1} K \text{diag}(q), \quad S = \text{diag}(q) K \text{diag}(q).$$

These identities give the restrictions. The full CES matrix proof is in the appendix. $\square$

The sign and reciprocity conditions have distinct content. Conditional on other outputs, an increase in one buyer’s production cannot lower another’s materials cost under the maintained substitutable-input supply structure. Expenditure weighting makes these marginal cost effects reciprocal and gives nonnegative aggregate curvature. Neither restriction requires the downstream matrix H to be symmetric. They apply standard cost duality and substitution arguments to the two contract-state responses; Galichon et al. (2025) develop the broader connection between substitution and inverse monotonicity, and Weyl and Fabinger (2013) provide the incidence framework. These are necessary conditions for the maintained competitive-supply model, not sufficient conditions for a conduct classification.

The contract comparison is part of the restriction. All active unit prices are inherited in F and all reset to common supplier marginal costs in R, at the same baseline and with outside quantities fixed. If only selected prices reset, buyers can face different prices from the same supplier and the proposition’s common-price condition need not hold. Applying the diagnostic therefore requires information about which contracts adjust as well as the corresponding output responses and input expenditures.

3.2 Local bargaining examples

Consider two symmetric Cournot buyers using one common supplier, with $\alpha_{i1} = z_i = c_1 = A = 1$, $Q_1^O = 0.4$, $(\rho, \sigma) = (4, 6)$, and $\gamma = 1/2$. With flat supplier marginal cost ($\psi_1 = 0$), the numerically validated Nash-in-Nash equilibrium gives

$$H = \begin{pmatrix} -4.375 & 0.375 \\ 0.375 & -4.375 \end{pmatrix}, \quad B \simeq \begin{pmatrix} -4.128833 & 0.128833 \\ 0.128833 & -4.128833 \end{pmatrix}, \quad (11)$$

and

$$N \simeq 0.012172 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad E_t = E_r \simeq 0.195325. \quad (12)$$

Own responses are negative and cross responses positive in both contract states, as in the competitive examples. Yet the negative cross entries of N violate Proposition 1. Here S is symmetric and positive semidefinite to numerical precision: the failure is specifically the cross-entry sign restriction. Price adjustment changes relative terms between the two buyers even when the supplier’s marginal cost is flat.

Two nearby specifications separate the restrictions further. Changing productivities to $(z_t, z_r) = (1.02, 0.98)$ gives $N_{tr} \simeq -0.011062$ and $N_{rt} \simeq -0.013157$, with $S_{tr} \simeq -0.002304667$ and $S_{rt} \simeq -0.002402719$; expenditure weighting no longer restores reciprocity. Keeping symmetry but setting $\psi_1 = 0.01$ leaves both cross entries near -0.010575 , so the sign violation also occurs with a positive supply slope. These are local numerical examples. Both shock columns, two derivative steps, and independent implementations of the two flat-cost cases support them; the appendix gives the checks. They do not impose a general sign on bargaining responses. In the HBM calibration, the relevant difference instead appears in weighted reciprocity (Table 2).

3.3 Downstream competition and policy incidence

The supply restriction and the downstream comparison use different objects. Under the independent-demand constant-markup specification, $H = -\nu I$: with input prices fixed, another buyer's wedge has no output effect. Cournot competition generally gives nonzero cross entries in H . In either demand system, resetting prices can transmit a shock through shared suppliers, and bargaining can violate the restrictions on N . Thus a violation of (10) alone does not distinguish NiN-Cournot from NiN-CM.

To interpret a targeted restriction, write $\theta = \log \tau_t$ and let $e_j = d \log \mu_j / d\theta$ along the reset path. Denote the fixed- and reset-price output elasticities by y^F and y^R .

Lemma 1 (Cost-vector response identity). *For differentiable paths through the same baseline and direct shock vector $v = d \log \tau / d\theta$,*

$$y^F = Hv, \quad y^R = H(v + e), \quad D := y^R - y^F = He. \quad (13)$$

For a shock to buyer t alone, $v_t = 1$ and $v_j = 0$ for $j \neq t$.

This chain rule retains the targeted buyer's cost relief as well as the rival's. The materials-cost derivative is the expenditure-share-weighted sum of active link-price derivatives.

Corollary 1 (Rival relief relative to target relief). *In the two-buyer Cournot model with $1 < \rho \leq \sigma$, $H_{rr} < 0$ and $H_{rt} + H_{rr} = -\rho$. Hence*

$$D_r = H_{rt}e_t - (\rho + H_{rt})e_r. \quad (14)$$

If $\sigma > \rho$, define $a = 1 - 1/\sigma$ and $\delta = 1/\rho - 1/\sigma$. Then $H_{rt} > 0$ precisely when

$$s_r < \frac{a}{a + \delta}. \quad (15)$$

In this region, if both buyers receive materials-cost relief $R_j = -e_j \geq 0$, resetting prices amplifies the rival's local output response if and only if

$$(\rho + H_{rt})R_r > H_{rt}R_t. \quad (16)$$

Equality leaves the response unchanged; reversing the inequality attenuates it. If $H_{rt} \geq 0$ and $e_t \leq 0 \leq e_r$, adjustment weakly attenuates the response. When $\sigma = \rho$, $H_{rt} = 0$ and $D_r = -\rho e_r$.

Target relief weakens the rival's competitive advantage, whereas rival relief expands its output. The share restriction matters because a rival that already serves almost the entire market may lose output as aggregate demand contracts, despite gaining market share. Appendix A develops shared and exclusive supply benchmarks that illustrate the two-cost condition. Finite policy effects require re-solving both contract states; the local identity does not by itself determine profit or welfare changes.

4 HBM evidence and quantitative design

4.1 Products, contracts, and physical supply

HBM sourcing motivates the network and contract states in the model. Micron’s March 2025 announcement places its 36GB 12-high HBM3E in NVIDIA B300 and GB300 products; its June announcement describes the same memory specification in AMD’s MI350 series, including compatibility work (Micron, 2025a,b). These disclosures establish a product-specific shared-supplier possibility. They leave contemporaneous bilateral purchases and the complete qualified network unreported. In particular, an absent announcement is not an observed zero link.

Contracts precede delivery. SK hynix described supply volumes as mutually agreed one year ahead in April 2025, while Micron’s June 2024 remarks stated that prices had been contracted for most 2025 HBM supply (SK hynix, 2025; Micron, 2024). This evidence motivates comparing inherited and adjustable prices. Advance quantity commitments also mean that the model’s freely adjusting orders under fixed unit prices are a specified contract state, not a state directly observed in these announcements. Neither F nor R is assigned to a particular elapsed time after a historical policy shock.

The quantitative source is the April 17, 2026 vintage of TrendForce’s HBM datasheet, using its historical industry estimates for calendar year 2025 (TrendForce, 2026). Generation and stacking-layer margins do not identify every joint product cell. We construct feasible HBM3e–12-high intersections from those margins and sum quarterly bounds to annual ranges, without multiplying shares or imposing unreported structural zeros. Table 1 reports the resulting supply ranges in million gigabits. A jointly selected midpoint supplies the model’s product-volume targets. The two focal buyers’ shares of those volumes and their allocation across suppliers remain scenario assumptions.

Table 1: Annual 2025 supply evidence and conditional sourcing requirements

Object	Lower	Midpoint	Upper
<i>A. Product supply completions (million Gb)</i>			
Samsung	2,215	2,320.0	2,425
SK hynix	8,355	8,570.0	8,785
Micron	3,010	3,022.5	3,035
<i>B. Conditional physical sourcing shares (%)</i>			
Samsung share of supply	15.78	16.68	17.59
Rival Samsung share, 80:20	78.91	83.38	87.93
90:10 with target Samsung link absent	Infeasible throughout the supply range		

Panel A uses feasible HBM3e 12-high intersections of industry-estimated generation and layer margins from TrendForce, *2Q26 HBM Datasheet*, April 17, 2026 vintage. These are completion ranges, not confidence intervals or directly observed bilateral volumes. Panel B imposes a common positive focal share m for all suppliers, target:rival input totals of 80:20 or 90:10, and zero target sourcing from Samsung. Its 78.91–87.93% interval is a requirement of those joint assumptions; it is not a measured buyer allocation. Midpoint means the jointly selected supplier-volume midpoint.

4.2 What the supply ranges require of a sparse allocation

Let S_u denote a feasible annual product volume and m_u the fraction allocated to the two focal buyers. Their total physical inputs are $F = \sum_u m_u S_u$, and the rival receives fraction b_r of F . If the target cannot source from Samsung, all Samsung focal volume must go to the rival. With both buyers otherwise able to source from SK hynix and Micron, nonnegative physical allocations exist if and only if

$$m_S S_S \leq b_r \sum_u m_u S_u. \quad (17)$$

This is an accounting feasibility condition; equilibrium prices and incentives impose further restrictions.

When all suppliers have a common positive focal share m , the required Samsung share of rival inputs is s_S/b_r , where $s_S = S_S/\sum_u S_u$. The annual supply ranges imply $s_S = 15.78$ – 17.59 percent. A target:rival allocation of 80:20 therefore requires the rival to obtain 78.91–87.93 percent of its physical inputs from Samsung, with 83.38 percent at the supplier-volume midpoint. The corresponding 90:10 sparse allocation is infeasible throughout the annual range. These statements concern the joint assumptions of common m , the buyer allocation, and the absent target–Samsung link. They do not estimate actual customer sourcing. Supplier-specific focal shares can relax the constraint; Appendix B.7 gives the thresholds.

Mapping supplier shipments into the model’s same-period production and inputs additionally assumes no inventory change or loss. The available accelerator-product panel does not resolve the required product, timing, and inventory bridge, so it provides no further numerical restriction on the 80:20 allocation. The appendix records a conditional reconciliation requirement. Broad customs unit values, retained in Appendix C, combine buyers and products and do not supply the bilateral prices or contract-state responses needed to apply the joint-response restrictions empirically.

4.3 Calibration targets and imposed primitives

Let $V = \sum_u S_u$ and $\bar{S}_u = S_u/V$ for the selected supplier-volume completion. Inputs and outside quantities use this normalized unit; multiplying by V returns million Gb. At the midpoint, $V = 13,912.5$ million Gb. For focal shares m_u and buyer physical input shares b_d , the common calibration quantity targets are

$$\sum_d x_{du} = m_u \bar{S}_u, \quad Q_u^O = (1 - m_u) \bar{S}_u, \quad \sum_u x_{du} = b_d \sum_u m_u \bar{S}_u. \quad (18)$$

These targets combine industry-estimated product margins, a selected completion, and imposed customer allocations. The physical demand equation is (2): the policy wedge is not multiplied into physical input demand. Input expenditure uses price times this same quantity.

The core calibration imposes $m_u = 0.5$, input elasticity $\eta = 2.5$, buyer bargaining weight $\gamma = 0.5$, and downstream parameters $\sigma = 6$, $\rho = 4$ (or $\nu = 6$ for independent constant-markup demand). Potential input weights are $1/3$ on every link. The sparse network removes the target’s Samsung link without renormalizing the surviving weights. Supplier cost curvature is indexed

by $\zeta_u = \psi_u \bar{S}_u$, set to 0.25 or zero. At the calibrated baseline, a proportional change in focal purchases changes marginal cost with elasticity $m_u \zeta_u / (1 + \zeta_u)$, holding outside quantities fixed.

Conditional on these inputs, two relative supplier cost scales and two buyer productivities match four independent physical margins: both buyer totals and two supplier focal totals. The third supplier margin follows by adding up. One supplier cost scale and aggregate demand are normalized to one. This is an exact conditional fit to calibration targets, and it leaves the imposed bargaining weight and elasticities unestimated. Final output and profit remain in model units. Table 5 lists the input status and Table 6 the fitted values.

4.4 Two comparisons of network structure

The first comparison holds economic primitives fixed. Starting from each saved dense or sparse 80:20 NiN–C calibration, it changes the link mask or the upstream pricing rule, keeping supplier cost scales and slopes, buyer productivities, potential input weights, outside quantities, and other parameters unchanged. Each resulting economy has its own newly solved baseline. F inherits that baseline’s prices; R permits them to adjust after the shock. Buyer shares, focal supplier shares, and deviations from the original quantity targets are outcomes in this exercise.

The second comparison refits the cost scales and productivities separately to the common quantity targets in each network and pricing specification. Five scenarios comprise the original design: dense 90:10 with positive curvature, dense and sparse 80:20 with positive curvature, and the corresponding flat-cost 80:20 pair. Each is fitted under MC–C, NiN–C, and NiN–CM. Comparing these fitted economies holds the quantity margins fixed while allowing primitives to change. Within every policy experiment, however, the baseline primitives remain fixed. The two network comparisons thus answer different questions: adjustment to a changed sourcing opportunity at fixed primitives, and incidence across alternative economies that match the same calibration targets.

The customer-allocation comparison extends the positive-curvature Cournot cases to target physical input shares $b_t = 0.75, 0.80, 0.82$. Supplier volumes, common focal shares $m_u = 0.5$, and imposed elasticities are unchanged; each network and pricing rule is fitted separately to the corresponding quantity margins. These are conditional allocations, not observed customer purchases or revenue shares. A separate sparse-network exercise combines the dense and sparse NiN–C cost and productivity blocks without refitting (Appendix B.4).

The joint-response calculations use both buyer shocks at each fitted baseline, while the finite policy exercise sets $\tau_t = 1.05$ and $\tau_r = 1$. The MC benchmark applies each supplier’s common marginal-cost price to its active buyers. Numerical acceptance, restarts, and the one unresolved fixed-primitive baseline are reported in Appendix B.

5 Contract adjustment in the HBM application

5.1 Joint responses distinguish restrictions that signs do not

The calibrated HBM examples make weighted reciprocity consequential. Table 2 constructs $N = H^{-1} - B^{-1}$ from both buyers' shocks at each specification's common baseline and weights its rows by input expenditure. In both dense and sparse sourcing, MC-C and NiN-C have positive cross entries of N . Unlike the small bargaining examples above, these HBM bargaining calibrations do not violate cross-entry nonnegativity. Their distinction from the maintained competitive benchmark lies in the joint relationship between the two directions of response.

Table 2: Joint-response restrictions at the HBM calibration targets

Network	Model	$\min(N_{tr}, N_{rt})$	S_{tr}	S_{rt}	$\max S - S^\top $
Dense	MC-C	0.020000	0.0122685	0.0122685	8.56×10^{-11}
Dense	NiN-C	0.022685	0.0171032	0.0120594	5.04×10^{-3}
Dense	NiN-CM	0.020316	0.0142965	0.0142968	3.46×10^{-7}
Sparse	MC-C	0.004581	0.0025996	0.0025996	3.00×10^{-11}
Sparse	NiN-C	0.005293	0.0035876	0.0022917	1.30×10^{-3}
Sparse	NiN-CM	0.004634	0.0029388	0.0030572	1.18×10^{-4}

$N = H^{-1} - B^{-1}$ and $S = \text{diag}(E)N$. Each row uses both buyer shocks at its own common baseline, with $\tau = 1$. Entries are local derivatives; S uses normalized model input expenditure. Competitive supply requires nonnegative N and symmetric positive-semidefinite S . All listed cross entries are positive. The two MC rows satisfy weighted reciprocity to numerical precision, whereas the NiN rows need not. Physical calibration targets are common; fitted primitives can differ across rows.

The distinction survives the three customer allocations $b_t = 0.75, 0.80, 0.82$ in both networks. All six NiN-C cases violate weighted reciprocity beyond the numerical sensitivity criterion; none of the six MC-C cases does. For $G = \text{diag}(E)^{1/2}N \text{diag}(E)^{-1/2}$, relative asymmetry $\|G - G^\top\|_F / (2\|G\|_F)$ ranges from 0.0876 to 0.1125 under dense NiN-C and from 0.0315 to 0.0367 under sparse NiN-C, compared with 4.8×10^{-10} – 2.1×10^{-9} under MC-C. All entries of N remain positive. These model-generated responses illustrate the additional information in weighted reciprocity when signs are shared; the asymmetry measure is neither a statistical test nor an estimate of bargaining power.

NiN-CM can also depart from the competitive restrictions, including weighted reciprocity. Its fixed-price matrix H is diagonal, but input-price adjustment can generate cross-buyer responses in B . The Cournot distinction therefore concerns the downstream cross-response in H ; a violation in N alone does not separate the two bargaining specifications. The calculations retain this distinction between the downstream strategic channel and the upstream restrictions.

5.2 A link change and a refitted network give different incidence

For a finite shock, define the rival's contract-adjustment contribution as

$$A_r(\tau_t) = \log q_r^R(\tau_t) - \log q_r^F(\tau_t). \quad (19)$$

The percentage-point counterpart is $100(q_r^R - q_r^F)/q_r^0$, where q_r^0 is that economy’s baseline output. Table 3 reports both total output responses and their percentage-point difference for NiN-C; the logarithmic contrasts are in the appendix.

Table 3: Contract adjustment with fixed primitives and with common targets

Network	Rival output response			Baseline quantity-target drift (%)		
	F (%)	R (%)	R-F (pp)	Target	Rival	Focal max.
<i>A. Dense NiN-C primitives held fixed</i>						
Dense	5.45	8.50	3.06	0.00	0.00	0.00
Sparse	3.61	4.90	1.29	-28.61	20.46	71.98
<i>B. Each network refitted to common quantity targets</i>						
Dense	5.45	8.50	3.06	0.00	0.00	0.00
Sparse	7.03	5.54	-1.48	0.00	0.00	0.00

NiN-C; five-percent targeted friction, $\tau_t = 1.05$. F and R inherit the same cell-specific baseline and differ only in price adjustment. Output percentages use that baseline output as denominator. Panel A changes only the link mask; supplier costs, buyer productivities, potential input weights, outside quantities, and other primitives remain fixed. Panel B refits relative cost scales and productivities to the same calibration quantity targets. Drift is $100(\text{baseline quantity}/\text{target} - 1)$; Focal max. is the largest absolute supplier focal-volume drift. Zero entries are rounded numerical residuals.

At the dense 80:20 calibration, the five-percent friction raises rival output by 5.45 percent under fixed prices and 8.50 percent under reset prices, an additional 3.06 percentage points. Removing the target’s Samsung link while retaining the dense primitives reduces the additional response to 1.29 points: F and R become 3.61 and 4.90 percent. Contract adjustment still amplifies rival expansion. The new no-shock economy no longer matches the original input totals: the target is 28.61 percent below its target and the rival 20.46 percent above, with a maximum supplier focal-volume deviation of 71.98 percent.

Refitting the sparse economy to the common quantity targets produces a different comparison. F is 7.03 percent and R is 5.54 percent, so contract adjustment contributes -1.48 points. Rival output still increases in both contract states. The additional effect changes sign because the fitted sparse economy also changes relative supplier costs, buyer productivities, and the resulting allocation of inputs. For example, Samsung supplies 19.39 percent of rival inputs after link removal at dense primitives, compared with 83.38 percent at the refitted sparse baseline. The negative contribution is therefore not a general implication of deleting the link.

The broader fixed-primitive comparison reinforces the dependence on the baseline economy. At dense NiN-C primitives, MC-C’s contribution falls from $+6.21$ to $+3.36$ points as the link is removed. At sparse NiN-C primitives, MC-C yields $+12.33$ points in the dense network and -0.98 in the sparse network. This latter dense economy departs substantially from the quantity targets. Seven of the eight planned cells have accepted baselines; the sparse-anchor, dense NiN-C cell remains unresolved (Table 13). The cost-productivity combinations in Appendix B.4 further show that neither block alone accounts for the sign change between the two sparse NiN-C economies.

5.3 Customer allocation disciplines the quantitative conclusion

Table 4 varies the target’s physical input share while retaining the supplier-volume targets. In the sparse network, Samsung’s share of rival inputs rises from 66.70 to 92.64 percent across the three allocations. NiN–C contract adjustment attenuates rival expansion at each of these points and amplifies it in the dense network. Sparse MC–C also attenuates expansion at 80:20 and 82:18; its contribution is positive at 75:25. The sign of attenuation therefore does not identify bargaining. Because cost scales and productivities are fitted separately, these comparisons assess alternative economies consistent with the specified quantity margins, rather than an isolated change in conduct.

Table 4: Customer allocation and the rival’s contract-adjustment response

Target input share (%)	Sparse rival’s Samsung share (%)	Dense: R–F (pp)		Sparse: R–F (pp)	
		MC–C	NiN–C	MC–C	NiN–C
75	66.70	+3.914	+2.645	+0.453	-0.302
80	83.38	+4.228	+3.056	-1.169	-1.482
82	92.64	+4.365	+3.263	-2.615	-2.568

Five-percent target cost wedge. R–F is $100(q_r^R - q_r^F)/q_r^0$ using each cell’s own baseline. Rival output rises under both F and R in every cell. Supplier-volume targets, $m_u = 0.5$, and $\zeta_u = 0.25$ are common; c and z are refitted for each allocation, network, and pricing rule. Samsung shares are conditional baseline physical allocations, not observed customer purchases; sparse shares coincide across pricing rules by quantity accounting. MC–C uses common marginal-cost prices on each supplier’s active links.

5.4 Why contract adjustment can attenuate rival expansion

The local decomposition $D = He$ explains how the refitted sparse result arises. In dense NiN–C, the rival’s own cost-relief channel contributes +0.855 to its output elasticity, while target cost relief contributes -0.202, giving +0.654. In the refitted sparse economy the corresponding terms are +0.041 and -0.337, giving -0.295. The restored competitiveness of the target outweighs the rival’s own cost relief. At the five-percent shock, both input-cost indices still fall in that sparse economy: the target’s by 1.083 log percent and the rival’s by 0.013 log percent. A falling rival input-cost index therefore need not imply that contract adjustment amplifies its output gain. Table 10 reports the derivatives separately from finite changes.

At $b_t = 0.82$, the sparse rival’s own input-cost index instead rises: by 0.067 log percent under MC–C and 0.118 under NiN–C at the five-percent shock. In the local NiN–C decomposition, target cost relief contributes -0.3931 and the rival’s own cost increase contributes -0.1138, summing to -0.5069. Both channels now attenuate rival expansion. The explanation based on two falling cost indices applies to the 80:20 calibration, while the full cost map also accommodates this change in the rival’s cost direction.

The supplementary scenarios clarify the channels. Flat supplier marginal costs eliminate the reset-minus-fixed effect under MC–C. With flat costs, the calibrated NiN–C contribution is negative in both networks, whereas NiN–CM’s is zero to numerical precision. With positive curvature, NiN–CM has a zero rival response under fixed prices and positive responses under reset prices. These comparisons distinguish common supply-cost adjustment from bargaining

and downstream competition within the specified economies. They do not make a negative contribution exclusive to bargaining: the refitted sparse MC-C contribution is also negative.

The HBM application illustrates why both contract-state responses and customer allocations matter. Joint-response restrictions distinguish the maintained supply mechanisms when individual signs coincide, while sourcing allocations govern the size and channels of incidence. The supply evidence constrains feasible quantities; bilateral product allocations and observed responses under identifiable contract states would be needed to apply this distinction to HBM conduct or policy magnitudes.

6 Conclusion

Comparing joint output responses under fixed and reset input prices exposes supply restrictions that individual response signs can conceal. At a common baseline, subtracting the inverse response matrices removes the common downstream response. Under the maintained competitive-supply assumptions, the resulting matrix is entrywise nonnegative and becomes symmetric positive semidefinite after input-expenditure weighting. Bargaining examples violate these necessary conditions even when the signs of individual output responses coincide. The contribution is a link between contract-state comparisons and restrictions on the supply mechanism that allows downstream strategic interaction.

The HBM application illustrates both the value and the information requirements of this comparison. Contract adjustment can attenuate rival expansion under competitive supply and bargaining, but the calibrated bargaining economies violate weighted reciprocity across the customer allocations examined. The magnitude of attenuation depends on sourcing allocations and on the primitives used to fit quantity targets. A link removal at fixed primitives therefore need not reproduce the result from a separately fitted sparse economy. Product supply quantities constrain feasible allocations; matched customer purchases and responses under known contract states would be needed to assess these restrictions in industry data.

A Joint-response restrictions and incidence benchmarks

A.1 Full proof of Proposition 1

All quantities in this proof are evaluated at the common baseline. Retain suppliers with positive focal sales, and set inactive-link inputs to zero. Define the buyer expenditure shares, supplier focal-sales shares, and marginal-cost elasticities by

$$W_{iu} = \frac{p_u x_{iu}}{E_i}, \quad T_{ui} = \frac{x_{iu}}{Q_u^F}, \quad \kappa_u = \frac{G_u''(Q_u^T)Q_u^F}{p_u} \geq 0. \quad (20)$$

Both W and T have unit row sums. For (4), $\kappa_u = c_u \psi_u Q_u^F / MC_u$. The elasticity uses focal volume because outside volume is fixed; no assumption of fixed outside expenditure is needed.

Write $\ell_u = d \log p_u$, $y_i = d \log q_i$, and $e_i = d \log \mu_i = d \log P_i^M$. The fixed-productivity CES expenditure identity implies $e = W\ell$. Differentiating physical input demand on active links and common marginal-cost pricing gives

$$d \log x_{iu} = y_i + \eta e_i - \eta \ell_u, \quad (21)$$

$$\ell_u = \kappa_u \sum_i T_{ui} (y_i + \eta e_i - \eta \ell_u). \quad (22)$$

There is no direct wedge term in physical input demand at fixed q , as specified in (2). Therefore, with

$$R = \text{diag} \left(\frac{\kappa_u}{1 + \eta \kappa_u} \right), \quad M = WRT, \quad (23)$$

the supply response satisfies

$$\ell = RT(y + \eta e), \quad e = M(y + \eta e), \quad e = Ly, \quad L = (I - \eta M)^{-1}M. \quad (24)$$

Each diagonal entry of R belongs to $[0, 1/\eta)$, so $M \geq 0$ and every row sum of M is strictly below $1/\eta$. Thus the spectral radius of ηM is less than one. The convergent expansion

$$L = \sum_{k=0}^{\infty} \eta^k M^{k+1}$$

proves entrywise nonnegativity, including suppliers with zero curvature.

Let $\Omega = \text{diag}(E)$. Common supplier prices imply

$$(\Omega M)_{ij} = \sum_u \frac{p_u R_{uu}}{Q_u^F} x_{iu} x_{ju}. \quad (25)$$

This is a symmetric positive-semidefinite Gram matrix. Since $E_i > 0$, the similar matrix $\widetilde{M} = \Omega^{1/2} M \Omega^{-1/2}$ is symmetric and positive semidefinite. Its eigenvalues satisfy $0 \leq m < 1/\eta$, so

$$\widetilde{L} = (I - \eta \widetilde{M})^{-1} \widetilde{M}$$

is symmetric and has eigenvalues $m/(1 - \eta m) \geq 0$. Consequently $\Omega L = \Omega^{1/2} \tilde{L} \Omega^{1/2}$ is symmetric and positive semidefinite.

Finally, let $v = d \log \boldsymbol{\tau}$. The downstream response is $y = H(v + e)$; on the reset path it is also $y = Bv$. Substituting $e = Ly$ gives

$$B^{-1} = H^{-1} - L, \quad N = L.$$

Thus $N \geq 0$ and ΩN is symmetric positive semidefinite. This argument uses no symmetry restriction on the downstream inverse response. Common η supplies the stated CES representation; it is a sufficient condition for this proof, not a claim that heterogeneous input elasticities must destroy the restrictions.

A.2 Verification of the local bargaining examples

The three Cournot examples in Section 3.2 use $\eta = 2.5$ and the stated one-supplier primitives. The first has $\psi_1 = 0$ and $(z_t, z_r) = (1, 1)$; the second changes only productivities to $(1.02, 0.98)$; the third changes only the slope to $\psi_1 = 0.01$. With a single supplier and $\alpha_{i1} = 1$, $P_i^M = p_{i1}$ and $x_{i1} = q_i/z_i$, so these physical input demands do not depend on η .

The saved `joint_response_results.json` computes both columns of H and B using separate buyer shocks, with central log-wedge steps 10^{-4} and 5×10^{-5} . The displayed values use the smaller step. Reset equilibria have positive bilateral gains, satisfy the Nash first-order conditions, and pass the recorded local-curvature and finite unilateral-price-deviation checks. The smallest singular value of each example's Nash-price Jacobian exceeds 26. These checks support the local numerical solutions; the finite deviation search does not establish global equilibrium uniqueness.

An independent implementation solves downstream quantities directly and constructs the bargaining first-order conditions with complex-step derivatives. For the symmetric and asymmetric flat-cost cases, its reset response matrices differ from the production implementation by less than 5×10^{-10} . At the smaller step in the symmetric flat-cost example, the eigenvalues of S are approximately 0 and 0.004755034, and the weighted symmetry error is below 3×10^{-12} . The asymmetric case has $|S_{tr} - S_{rt}| \simeq 9.80519 \times 10^{-5}$ at both derivative steps. The positive-curvature example has negative cross entries of N at both steps. These computations distinguish the sign and reciprocity failures without extrapolating them to all bargaining parameters.

A.3 Proof of Lemma 1

On each path, $\log C_j = \log \tau_j + \log P_j^M - \log z_j$. The two paths share the same baseline and therefore the same downstream derivative matrix H . Under inherited prices the fixed technology gives $d \log(P_j^M/z_j) = 0$, whereas under reset prices its derivative is e_j . Differentiating $\log q(C)$ yields $y^F = Hv$ and $y^R = H(v + e)$. Subtraction proves the result. For weighted CES, the usual chain rule and expenditure identity give

$$e_j = \sum_u \omega_{ju} \frac{d \log p_{ju}}{d\theta}, \quad \omega_{ju} = \frac{\alpha_{ju} p_{ju}^{1-\eta}}{\sum_v \alpha_{jv} p_{jv}^{1-\eta}}.$$

Weights and productivity are held fixed as primitives, not recomputed by row normalization after a link change. Across finite shocks, the two paths generally have different H matrices away from the baseline. Thus He at the baseline is a local contrast, not an exact finite-change formula.

A.4 Proof of Corollary 1

Let $a = 1 - 1/\sigma$, $\delta = 1/\rho - 1/\sigma \geq 0$, and $a_i = a - \delta s_i > 0$. The Cournot first-order condition is $C_i = p_i^D a_i$. Since

$$\frac{\partial \log p_i^D}{\partial \log q_j} = -\frac{\mathbf{1}\{i = j\}}{\sigma} - \delta s_j, \quad \frac{\partial s_i}{\partial \log q_j} = a s_i (\mathbf{1}\{i = j\} - s_j),$$

the cost-to-output inverse Jacobian $J_{ij} = \partial \log C_i / \partial \log q_j$ has entries

$$J_{ii} = -\frac{1}{\sigma} - \delta s_i - \frac{\delta a s_i (1 - s_i)}{a_i} < 0, \quad (26)$$

$$J_{ij} = \frac{\delta s_j \{(a + \delta) s_i - a\}}{a_i}, \quad i \neq j. \quad (27)$$

Both row sums are $-1/\rho$. In two dimensions, one eigenvalue is $-1/\rho$ and the other is

$$-\frac{1}{\sigma} - \delta a s_t s_r \left(\frac{1}{a_t} + \frac{1}{a_r} \right) < 0.$$

Therefore $\det J > 0$, $H = J^{-1}$, and $H_{rr} = J_{tt} / \det J < 0$. Also $H(1, 1)' = -\rho(1, 1)'$. For $\delta > 0$, $H_{rt} = -J_{rt} / \det J$ is positive exactly when $s_r < a/(a + \delta)$. The threshold lies strictly between one half and one because $0 < \delta < a$. If $\delta = 0$, $H = -\rho I$. Substitution into $D_r = H_{rt} e_t + H_{rr} e_r$ proves all claims. This derivative calculation uses the stated two-buyer demand system; it is not a general Cournot cross-output sign theorem.

A.5 A shared-exclusive marginal-cost benchmark

Proposition 2 (Opposite link responses with rival cost relief under MC pricing). *Suppose two symmetric buyers have independent demand with elasticity $\nu > \eta > 1$. Each buys from a shared supplier S and its own exclusive supplier E_d . At a symmetric interior baseline, let $w \in (0, 1)$ be either buyer's expenditure share on S . Suppliers have common marginal costs across their customers, and prices reset to marginal cost. Let $\kappa_S, \kappa_E > 0$ be the elasticities of marginal cost with respect to focal supplier volume, holding outside volume fixed. For $\theta = \log \tau_t$, write $\ell_S, \ell_{Et}, \ell_{Er}$ for the reset log-price derivatives. Define*

$$L = 1 + \kappa_E \{\eta w + \nu(1 - w)\}, \quad \beta = \frac{w(1 + \kappa_E \eta)}{L}.$$

Then

$$\ell_S = -\frac{\kappa_S \nu (1 + \kappa_E \eta)}{2L[1 + \kappa_S\{\eta + (\nu - \eta)\beta\}]} < 0, \quad (28)$$

$$\ell_{Er} = \frac{\kappa_E(\eta - \nu)w\ell_S}{L} > 0, \quad e_r = \beta\ell_S < 0, \quad D_r = -\nu e_r > 0. \quad (29)$$

Proof. Independent demand and fixed productivity imply $d \log q_d / d\theta = -\nu(v_d + e_d)$ with $v_t = 1, v_r = 0$. Differentiating physical CES input demand gives

$$\frac{d \log x_{du}}{d\theta} = -\nu v_d + (\eta - \nu)e_d - \eta \ell_u.$$

Exclusive-supplier MC pricing therefore satisfies

$$\ell_{Ed} = \kappa_E[-\nu v_d + (\eta - \nu)e_d - \eta \ell_{Ed}], \quad e_d = w\ell_S + (1 - w)\ell_{Ed}.$$

Solving yields $\ell_{Ed} = \kappa_E[-\nu v_d + (\eta - \nu)w\ell_S]/L$, hence $e_r = \beta\ell_S$ and $e_t = e_r - (1 - w)\kappa_E\nu/L$. Symmetry gives equal physical shares of the common supplier's focal sales, so

$$\ell_S = \kappa_S[-\nu/2 + (\eta - \nu)(e_t + e_r)/2 - \eta \ell_S].$$

Substitution produces (28). Every denominator is positive. Since $\nu > \eta$ and $\ell_S < 0$, the rival's exclusive price rises, but its aggregate index falls. Its inherited-price output is unchanged under independent demand, so $D_r = -\nu e_r > 0$. $\square$

For the quadratic cost curve in (4),

$$\kappa_u = \frac{\psi_u Q_u^F}{1 + \psi_u Q_u^T} = \frac{m_u \zeta_u}{1 + \zeta_u}, \quad m_u = \frac{Q_u^F}{Q_u^T}, \quad \zeta_u = \psi_u Q_u^T.$$

Thus positive curvature and positive focal coverage supply a nonempty domain under the stated common floors, fixed outside quantities, symmetry, and independent downstream demand.

For comparison, if symmetric Cournot buyers use only one common supplier, its price derivative is $\ell = -\kappa\rho/[2(1 + \kappa\rho)]$. Homogeneity gives the mean quantity derivative $-\rho/2 - \rho\ell$ and the MC condition is $\ell = \kappa(-\rho/2 - \rho\ell)$. Since both buyers receive the same input-cost change, $D_r = -\rho\ell = \rho^2\kappa/[2(1 + \kappa\rho)] > 0$. This additional formula does not require $\sigma = \rho$.

A.6 A common-supplier linear-bargaining benchmark

Proposition 3 (A nonempty NiN domain for amplification). *Let two symmetric buyers use one common supplier with unit input coefficients and independent downstream demand of elasticity $\nu > 1$. Write focal input quantities as $X_d = K\tau_d^{-\nu}p_d^{-\nu}$, $K > 0$. Supplier marginal cost is $a + b(Q^O + X_t + X_r)$ with $a, b > 0$ and fixed $Q^O \geq 0$. Under the model's linear-price Nash product, set*

$$A_\gamma = \frac{\nu - \gamma}{\nu - 1}, \quad B_\gamma = A_\gamma - \frac{\gamma}{2}, \quad k = \frac{bX}{p}$$

at the symmetric baseline. If $\nu k \gamma / 2 < 1$, the symmetric price system is locally regular and

$$D_r = \frac{\nu^2 k A_\gamma}{[1 + \nu k (2A_\gamma - \gamma/2)](1 - \nu k \gamma / 2)} > 0. \quad (30)$$

In particular, $\nu = 6$ and $\gamma = 1/2$ satisfy this regularity condition for every positive a, b, K and nonnegative Q^O .

Proof. Removing the sole input link makes that buyer exit. Independent demand leaves the other buyer's quantity unchanged. Buyer surplus is proportional to $p_d^{1-\nu}$; supplier surplus is

$$G_d^U = [p_d - h_d - \frac{1}{2}bX_d]X_d, \quad h_d = a + bQ^O + bX_{-d}.$$

Differentiating the log Nash product gives the price equation

$$p_d = A_\gamma(a + bQ^O) + b[B_\gamma X_d + A_\gamma X_{-d}]. \quad (31)$$

For fixed other price, the left side minus the right side is strictly increasing in p_d , from a negative to a positive value. At its unique root, $p_d - h_d - bX_d/2 = (A_\gamma - 1)h_d + (B_\gamma - 1/2)bX_d > 0$. The Nash product vanishes at the participation boundary and as p_d tends to infinity. The root is therefore the unique positive-surplus one-link maximizer, not merely an unchecked stationary point.

The symmetric equation $p = A_\gamma(a + bQ^O) + (A_\gamma + B_\gamma)bKp^{-\nu}$ has a unique positive solution. Its local log-price response obeys

$$\begin{pmatrix} 1 + \nu k B_\gamma & \nu k A_\gamma \\ \nu k A_\gamma & 1 + \nu k B_\gamma \end{pmatrix} \begin{pmatrix} \ell_t \\ \ell_r \end{pmatrix} = -\nu k \begin{pmatrix} B_\gamma \\ A_\gamma \end{pmatrix}.$$

The determinant is the denominator in (30). Inversion gives $\ell_r = -\nu k A_\gamma / \det < 0$ and $D_r = -\nu \ell_r$. Moreover $k < 1/(A_\gamma + B_\gamma)$. At $\nu = 6, \gamma = 1/2, A_\gamma = 11/10, B_\gamma = 17/20$, and $\nu k \gamma / 2 < 10/13 < 1$. This proves a nonempty domain and local regularity, without asserting uniqueness of every possibly asymmetric joint equilibrium. $\square$

The engine's Cournot demand coincides with this independent-demand benchmark at $\rho = \sigma = \nu$. Strict response inequalities persist locally under small demand-parameter changes along a regular admissible branch. A numerical example at $(\rho, \sigma) = (5.9, 6)$ verifies such a nearby case; it does not establish the signs at the principal $(4, 6)$ calibration or at arbitrary multisupplier networks.

A.7 Numerical checks of the incidence benchmarks

The standalone script `theory/verify_contract_reset.py` leaves the shared equilibrium engine unchanged. It checks the analytical formulas by exact rational substitutions and compares (13) with symmetric re-solves at log shocks 10^{-4} and 5×10^{-5} . The downstream H is independently measured by cost perturbations with input prices fixed. Seven small fixtures satisfy the checked

demand and price conditions; bargaining baselines also pass bilateral-gain and unilateral-price-deviation checks. These are uncalibrated examples, not data fits.

For the sparse examples the two buyers each have one common and one exclusive supplier, active CES weights are $1/2$, supplier scales and buyer productivities equal one, supplier slopes equal 0.3 , outside volumes are $(0.4, 0.2, 0.2)$, and $(\eta, \gamma) = (2.5, 0.5)$. At $\rho = \sigma = 6$, both pricing rules exhibit a lower rival shared price, a higher exclusive price, and $D_r > 0$. At $(\rho, \sigma) = (4, 6)$, the MC example has $D_r = 0.09526$, whereas the NiN example has $D_r = -0.18104$. In the latter, the rival’s shared and exclusive price derivatives are both positive, so the premise of opposite link signs fails. Pricing rules have different baseline equilibria; this comparison is a witness of conditionality, not a bargaining treatment effect. The full derivatives, audit results, and engine hash are recorded in `theory/verification.json`.

The analytical benchmarks establish nonempty domains for the two-cost incidence comparison and show why amplification and opposite link-price signs can occur under either pricing rule.

B Numerical implementation and supplementary outcomes

B.1 Calibration inputs and accepted core solutions

The replication uses the unchanged physical observation operator and equilibrium engine. The original quantity-target design records five scenarios, source and code hashes, allocation-feasibility witnesses, every baseline attempt, and every policy cell. Table 5 distinguishes the input assumptions from fitted quantities. The principal artifact contains 15 accepted baseline cells and 60 accepted policy cells; four local diagnostics and eight target-reversal cells are recorded separately. A 30-cell zero-shock check recomputes both contract states at every baseline, reproducing prices and quantities with maximum log error below 6×10^{-15} .

Table 5: Status of the calibration inputs

Object	Core setting	Status
Supplier physical supply	Table 1, midpoint	Selected completion of estimated product margins
Buyer input totals $b_t : b_r$	90:10 or 80:20	Hypothetical allocation
Focal supplier share m_u	0.5 for each supplier	Outside-customer scenario
Qualified links and α_{du}	Dense or specified sparse; potential weights $1/3$	Fixed scenario technology
Input elasticity η	2.5	Imposed
Downstream elasticities	$\sigma = 6$, $\rho = 4$; $\nu = 6$ in CM	Imposed demand specifications
Buyer bargaining weight γ	0.5	Imposed, not estimated
Cost slope index $\zeta_u = \psi_u \bar{S}_u$	0.25 or 0	Imposed curvature
Supplier costs and buyer productivity	Two relative c_u and two z_d	Fitted conditional on the above
Demand and price units	$A = 1$, one supplier’s $c_u = 1$	Normalizations

The fitting map has four free variables. All 15 baselines pass the physical-margin, equilibrium, and restart-stability requirements. Of 75 attempts, 63 are accepted and 12 rejected; rejected

attempts remain in the replication files. At least two accepted starts agree within 10^{-5} in maximum log-parameter dispersion. Accepted starts have full-rank local numerical fitting Jacobians, a diagnostic conditional on the imposed parameters. The equilibrium audit checks demand first-order conditions, price conditions, bilateral gains, and finite unilateral price deviations where applicable. These checks do not establish global uniqueness of the multisupplier bargaining equilibrium.

Table 6: Conditional baseline parameters and accepted starts

Scenario	Model	c_1	c_3	z_t	z_r	Starts
C0	MC-C	1.687	1.517	2.129	1.237	5/5
C0	NiN-C	1.626	1.471	2.790	1.551	2/5
C0	NiN-CM	1.592	1.447	2.233	1.466	5/5
C1	MC-C	1.687	1.517	2.049	1.435	5/5
C1	NiN-C	1.622	1.468	2.694	1.819	2/5
C1	NiN-CM	1.592	1.447	2.186	1.680	5/5
C2	MC-C	0.465	1.517	2.372	0.675	5/5
C2	NiN-C	0.515	1.459	3.007	0.900	2/5
C2	NiN-CM	0.514	1.441	2.482	0.876	5/5
C3	MC-C	1.687	1.517	1.521	1.066	5/5
C3	NiN-C	1.621	1.467	2.036	1.358	4/5
C3	NiN-CM	1.593	1.448	1.705	1.292	5/5
C4	MC-C	0.465	1.517	1.762	0.501	5/5
C4	NiN-C	0.505	1.460	2.288	0.670	3/5
C4	NiN-CM	0.504	1.443	1.945	0.671	5/5

$c_2 = 1$ is the price normalization. There are five starts per cell. Rejected attempts remain in the replication output. Differences in fitted parameters preclude interpreting cross-model response differences as a pure bargaining treatment effect.

B.2 Complete finite policy outcomes and local decomposition

Tables 7 and 8 report the five- and one-percent targeted shocks for the five scenarios. Table 9 includes the input-cost indices and targeted buyer’s output. No parameters are refitted after a shock. In the dense 90:10 case, the additional rival output response is 4.99 percentage points under MC-C and 4.43 under NiN-C; the reset responses are 12.41 and 11.94 percent. Thus substantial additional expansion is shared by both pricing rules.

The local steps are 10^{-4} and 5×10^{-5} in $\log \tau_t$. The analytic downstream matrix is checked by independently perturbing each buyer’s costs with input prices fixed. The complete local response identities agree to below 7×10^{-9} . Table 10 and Figure 1 separate these derivatives from the finite-shock log contrasts. The one-percent comparisons preserve the principal dense-sparse signs. A separate eight-cell experiment restricts the initially smaller buyer in the same four fitted Cournot baselines. Contract adjustment then amplifies the unrestricted larger buyer’s response in all four cases, by 0.24 points under sparse MC-C and 0.30 under sparse NiN-C. Which buyer is restricted matters without changing the calibration.

Table 11 reports Cournot revenue-share and total supplier-volume responses under reset prices. Table 12 reports operating-profit changes in normalized units, including suppliers’

Table 7: Rival output after a five-percent targeted cost friction

Scenario	Model	F output (%)	R output (%)	R-F (pp)	$100A_r$
C0	MC-C	7.42	12.41	4.99	4.54
C0	NiN-C	7.51	11.94	4.43	4.04
C0	NiN-CM	0.00	10.48	10.48	9.97
C1	MC-C	5.35	9.58	4.23	3.93
C1	NiN-C	5.45	8.50	3.06	2.86
C1	NiN-CM	0.00	9.26	9.26	8.85
C2	MC-C	7.12	5.95	-1.17	-1.10
C2	NiN-C	7.03	5.54	-1.48	-1.39
C2	NiN-CM	0.00	3.02	3.02	2.98
C3	MC-C	5.35	5.35	0.00	0.00
C3	NiN-C	5.48	3.75	-1.73	-1.65
C3	NiN-CM	0.00	0.00	0.00	0.00
C4	MC-C	7.12	7.12	0.00	0.00
C4	NiN-C	7.07	5.76	-1.30	-1.23
C4	NiN-CM	0.00	0.00	0.00	0.00

C0: dense 90:10; C1/C2: dense/sparse 80:20; C3/C4: the corresponding flat-cost cases. F and R denote fixed and reset unit prices. Output changes are percentages relative to each fitted baseline. R-F is a percentage-point difference; $100A_r = 100 \log(q_r^R/q_r^F)$ is a log contrast. These are different units.

Table 8: Rival output after a one-percent targeted cost friction

Scenario	Model	F output (%)	R output (%)	R-F (pp)	$100A_r$
C0	MC-C	1.51	2.50	0.99	0.97
C0	NiN-C	1.52	2.43	0.91	0.89
C0	NiN-CM	0.00	2.17	2.17	2.15
C1	MC-C	1.11	1.97	0.86	0.84
C1	NiN-C	1.13	1.78	0.65	0.64
C1	NiN-CM	0.00	1.94	1.94	1.93
C2	MC-C	1.45	1.21	-0.24	-0.24
C2	NiN-C	1.43	1.14	-0.30	-0.29
C2	NiN-CM	0.00	0.64	0.64	0.63
C3	MC-C	1.11	1.11	0.00	0.00
C3	NiN-C	1.14	0.80	-0.34	-0.34
C3	NiN-CM	0.00	0.00	0.00	0.00
C4	MC-C	1.45	1.45	0.00	0.00
C4	NiN-C	1.44	1.19	-0.25	-0.24
C4	NiN-CM	0.00	0.00	0.00	0.00

C0: dense 90:10; C1/C2: dense/sparse 80:20; C3/C4: the corresponding flat-cost cases. F and R denote fixed and reset unit prices. Output changes are percentages relative to each fitted baseline. R-F is a percentage-point difference; $100A_r = 100 \log(q_r^R/q_r^F)$ is a log contrast. These are different units.

Table 9: The two buyers' input-cost changes under reset prices

Scenario	Model	Target PM	Rival PM	Target output (%)
C0	MC-C	-1.135	-1.135	-13.92
C0	NiN-C	-0.887	-0.977	-14.77
C0	NiN-CM	-1.323	-1.662	-19.22
C1	MC-C	-0.984	-0.984	-14.63
C1	NiN-C	-0.811	-0.734	-15.20
C1	NiN-CM	-1.137	-1.476	-20.11
C2	MC-C	-1.266	-0.119	-13.48
C2	NiN-C	-1.083	-0.013	-14.11
C2	NiN-CM	-1.441	-0.496	-18.64
C3	MC-C	0.000	0.000	-17.93
C3	NiN-C	0.046	0.337	-18.05
C3	NiN-CM	0.000	0.000	-25.38
C4	MC-C	0.000	0.000	-17.75
C4	NiN-C	-0.004	0.228	-17.74
C4	NiN-CM	0.000	0.000	-25.38

The shock is five percent. PM entries are $100\Delta \log P^M$; the targeted buyer also faces the effective-cost wedge $100\log(1.05)$. Productivity is fixed within each experiment.

Table 10: Local output contribution of target relief and rival relief

Scenario	Model	e_t	e_r	$H_{rt}e_t$	$H_{rr}e_r$	D_r
C1	MC-C	-0.215	-0.215	-0.242	1.100	0.859
C1	NiN-C	-0.176	-0.166	-0.202	0.855	0.654
C2	MC-C	-0.272	-0.028	-0.397	0.151	-0.246
C2	NiN-C	-0.234	-0.008	-0.337	0.041	-0.295

All entries are derivatives with respect to $\log \tau_t$, evaluated at baseline. Cost responses e exclude the direct policy wedge. The last two terms sum to D_r , the reset-minus-fixed rival output elasticity. They are not percentage-point effects of the five-percent shock.

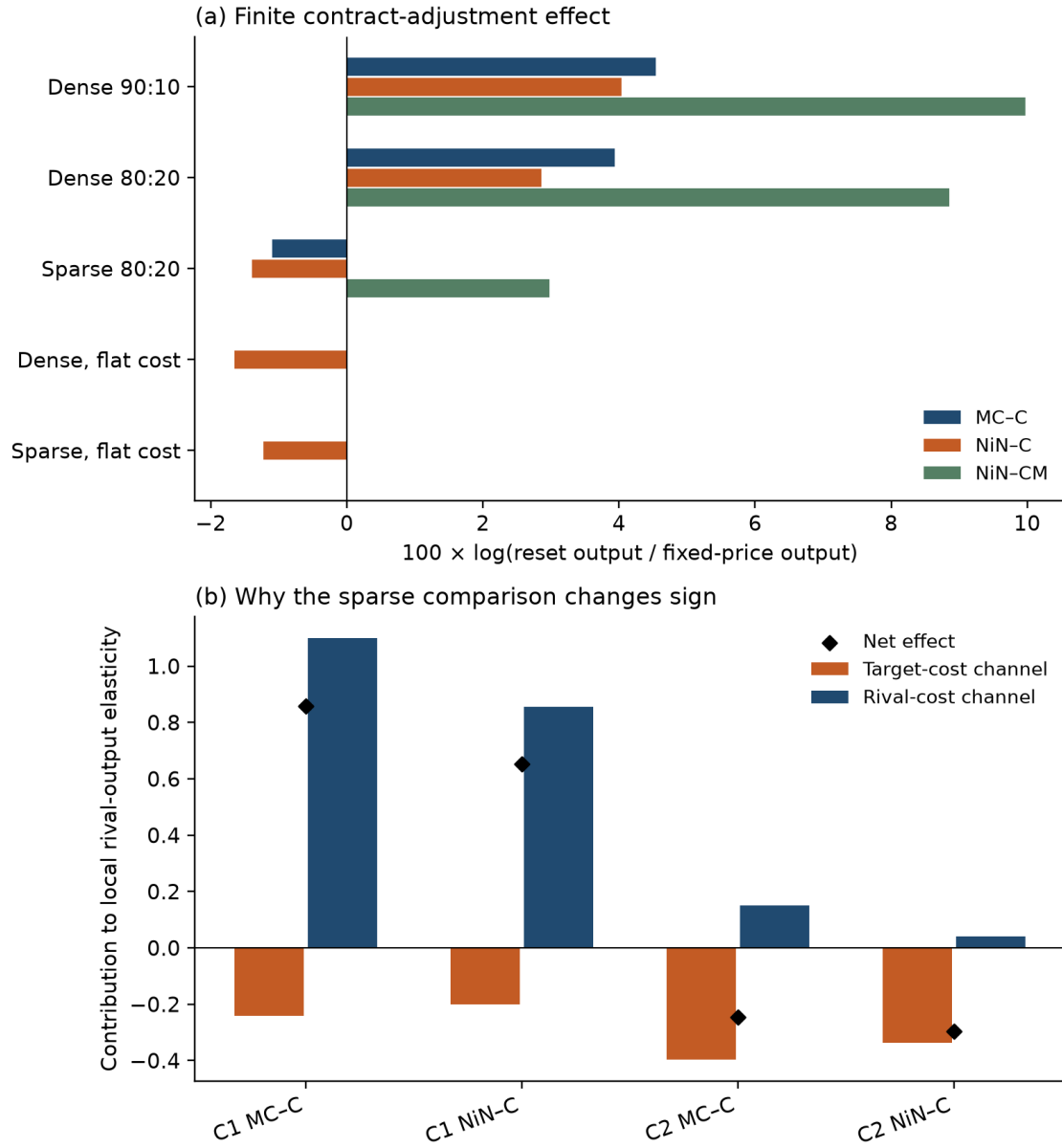


Figure 1: Supplementary contract-adjustment outcomes. Panel (a) reports the finite five-percent-shock log contrast $100A_r$ across five calibrated scenarios. Panel (b) reports local derivative contributions in the dense and sparse 80:20 Cournot cases. The panels have different units.

focal incremental profits. The accompanying `complete_policy_outcomes.csv` contains output, shares, effective and material costs, supplier volumes, and profit changes for all 60 core policy cells. Negative focal profits are flagged rather than causing an automatic price reset; no core cell has such a flag. Outside-customer profits are omitted.

Table 11: Rival revenue share and supplier total-volume responses

Scenario	Model	Rival share (pp)	Supplier 1	Supplier 2	Supplier 3
C0	MC-C	2.04	-5.64	-5.64	-5.64
C0	NiN-C	2.02	-6.08	-6.03	-6.08
C1	MC-C	3.41	-4.89	-4.89	-4.89
C1	NiN-C	3.29	-5.29	-5.20	-5.28
C2	MC-C	1.69	2.23	-6.29	-6.29
C2	NiN-C	1.78	2.02	-6.60	-6.60
C3	MC-C	3.41	-6.64	-6.64	-6.64
C3	NiN-C	3.12	-6.89	-6.82	-6.88
C4	MC-C	2.25	3.56	-8.38	-8.38
C4	NiN-C	2.17	2.93	-8.40	-8.42

Five-percent friction with reset prices. Supplier columns are total physical-volume changes in percent, including fixed outside quantities. The supplier order is Samsung, SK hynix, Micron; names label common supply targets rather than observed bilateral behavior. Share changes use model revenues.

Table 12: Operating-profit changes within each Cournot specification

Scenario	Model	Buyer t	Buyer r	Supplier 1	Supplier 2	Supplier 3
C0	MC-C	-25.296	2.018	-1.872	-4.100	-2.194
C0	NiN-C	-32.790	2.308	-7.422	-14.689	-8.567
C1	MC-C	-23.711	3.621	-1.636	-3.584	-1.918
C1	NiN-C	-30.106	3.932	-6.574	-13.026	-7.592
C2	MC-C	-19.814	1.051	0.221	-4.538	-2.428
C2	NiN-C	-24.760	1.265	0.620	-14.492	-8.778
C3	MC-C	-22.782	2.016	0.000	0.000	0.000
C3	NiN-C	-28.404	1.927	-5.306	-10.089	-6.092
C4	MC-C	-20.968	1.044	0.000	0.000	0.000
C4	NiN-C	-25.408	1.122	0.463	-11.161	-6.999

Five-percent friction with reset prices. Entries are 1000 times changes in normalized model profit, not dollars. Upstream entries cover focal incremental operating profit; outside-customer profit is omitted. Compare signs within a fitted economy, not currency levels across models. Full CM and fixed-price outcomes are in the accompanying CSV.

B.3 Fixed-primitive outcomes, target drift, and the unresolved cell

Both anchors are saved NiN-C calibrations: dense and sparse 80:20. Potential input weights come from the saved common design, with all six entries equal to $1/3$. Removing the target-Samsung link sets that weight to zero and leaves all surviving weights unchanged. Holding primitives fixed includes c, z, ψ, Q^O , cost multipliers, demand parameters, bargaining weights, and demand scale. The descriptors b, m, ζ are recomputed at the new baseline. Tables 13 and 14 report all eight planned cells rather than dropping the unresolved one.

The production root routine uses `hybr`, tolerance 10^{-10} , and a maximum of 700 function evaluations. It checks the inherited-price start and starts at 1.25, 1.8, and 2.5 times cost floors where applicable. Seven baselines and their policy and local states pass the saved certificates.

Table 13: All eight fixed-primitive comparison cells

Anchor	Network	Model	F (%)	R (%)	R-F (pp)	$100 \log(R/F)$
Dense	Dense	MC-C	5.7911	12.0047	6.2136	5.7075
Dense	Dense	NiN-C	5.4492	8.5049	3.0557	2.8566
Dense	Sparse	MC-C	3.4045	6.7598	3.3553	3.1933
Dense	Sparse	NiN-C	3.6100	4.8990	1.2890	1.2364
Sparse	Dense	MC-C	10.1011	22.4277	12.3266	10.6122
Sparse	Dense	NiN-C	Unresolved baseline; policy states not run			
Sparse	Sparse	MC-C	7.2080	6.2302	-0.9777	-0.9162
Sparse	Sparse	NiN-C	7.0250	5.5427	-1.4824	-1.3947

Each anchor is a saved NiN-C calibration. Seven of eight baselines were accepted. F and R refer to rival output under a five-percent targeted friction; percentages are relative to the corresponding baseline. The unresolved cell is retained and does not establish equilibrium nonexistence.

Table 14: Departure from the original quantity targets under fixed primitives

Anchor	Network	Model	Target	Rival	Focal max.	Total max.	100 <i>b</i> _{<i>t</i>}
Dense	Dense	MC-C	80.67	54.57	83.35	41.68	82.38
Dense	Dense	NiN-C	0.00	0.00	0.00	0.00	80.00
Dense	Sparse	MC-C	19.99	120.77	63.78	31.89	68.49
Dense	Sparse	NiN-C	-28.61	20.46	71.98	35.99	70.33
Sparse	Dense	MC-C	352.76	-91.85	959.80	479.90	99.55
Sparse	Dense	NiN-C	Unresolved baseline				
Sparse	Sparse	MC-C	65.22	45.43	74.25	37.13	81.96
Sparse	Sparse	NiN-C	0.00	0.00	0.00	0.00	80.00

The first four numeric columns are percentage deviations of baseline physical inputs from the original targets; supplier columns are maxima of absolute deviations. The last column is the new target-buyer physical input share in percent. These endogenous descriptors are not reimposed after changing the network or pricing rule.

Across the 84 accepted baseline, finite-policy, and local states, the maximum analytic first-order-condition residual is 4.7×10^{-13} . The two local steps agree to below 10^{-8} . Bargaining certificates require positive gains and local price maxima, with finite best-response scans over prices from 0.01 to 100 times the candidate price. Fixed-price states enforce inherited prices, physical accounting, and downstream equilibrium; they do not impose renegotiated Nash first-order conditions.

The sparse-anchor, dense NiN-C baseline fails at all four initial starts. An additional start at 1.25 times the accepted MC prices has positive gains but fails with residual 0.01070. A bounded continuation restores the missing potential weight through 0.005, 0.025, 0.08, 0.17, 1/3. The first two steps pass the full certificate; the third fails with residual 0.38404, and the remaining steps are not attempted. No failed iterate is treated as an equilibrium, and its policy and local outcomes are not computed. This limited numerical failure does not establish nonexistence. The accepted intermediate weights are continuation points, not additional economic comparison cells.

B.4 Cost and productivity blocks in the sparse network

Table 15 holds the sparse network and all primitives other than supplier cost scales c and buyer productivities z fixed. Each block comes from either the dense or sparse 80:20 NiN-C calibration. The dense-dense combination is the fixed-primitive link-removal economy; the sparse-sparse combination is the refitted sparse baseline. Mixed combinations are not fitted to the quantity targets. Since $MC_u = c_u(1 + \psi_u Q_u^T)$, changing c changes both cost levels and slopes $c_u \psi_u$.

Table 15: Cost and productivity combinations with the sparse network fixed

Cost c	Productivity z	F (%)	R (%)	R-F (pp)	b_t (%)	Samsung (%)	Drift (%)
Dense	Dense	3.610	4.899	+1.289	70.33	19.39	71.98
Sparse	Dense	1.033	1.707	+0.674	33.89	71.34	358.16
Dense	Sparse	9.821	13.901	+4.080	98.74	20.04	98.77
Sparse	Sparse	7.025	5.543	-1.482	80.00	83.38	0.00

NiN-C; five-percent targeted wedge. Dense and Sparse denote the source of each block in the saved 80:20 calibrations. Other primitives and the sparse link mask are fixed; each combination uses its own baseline without refitting. b_t is the resulting target physical input share; Samsung is its share of rival physical inputs. Drift is the maximum absolute supplier focal-volume deviation from the original 80:20 quantity targets. The zero is a rounded residual. These are model-implied quantities.

Replacing only costs or only productivities leaves the additional rival response positive, at +0.674 and +4.080 percentage points, respectively; replacing both gives -1.482 points. The four-cell interaction, $\Delta_{11} - \Delta_{10} - \Delta_{01} + \Delta_{00}$, is -4.947 points, where the first index selects the cost block and the second selects productivity (0 for dense, 1 for sparse). Costs contribute -0.615 points if changed first and -5.562 if changed second; productivities contribute +2.791 if changed first and -2.156 if changed second. Averaging the two orders gives -3.089 and +0.318 points, which sum to the endpoint change of -2.771 points. This is conditional block accounting, not a unique causal attribution.

The intermediate economies illustrate the interaction at the cost of substantial quantity drift. The target's actual physical input share is 33.89 percent with only the cost block replaced and 98.74 percent with only productivity replaced; the largest absolute supplier focal-volume deviations are 358.16 and 98.77 percent. These combinations dissect the refitting mechanism and are not additional fits to the industry quantity margins.

B.5 Customer-allocation and joint-response verification

The saved allocation extension contains 12 cells: three physical input allocations, two networks, and two Cournot pricing rules. Together with the four sparse cost-productivity combinations, it contains 16 logical cells and 15 distinct economies because the sparse-sparse endpoint also appears at $b_t = 0.80$. All recorded baselines, 32 finite policy states, and 16 joint-response cases pass the saved certificates. These counts include reused states. The extension leaves the unresolved fixed-primitive cell described above unresolved.

Both buyer shocks use the same baseline within each case. The two symmetric log-wedge steps are 10^{-4} and 5×10^{-5} ; reset responses are also checked against implicit differentiation. A weighted-reciprocity violation is classified as numerically resolved only when $\|G - G^\top\|_F$ exceeds ten times the saved sensitivity measure, which combines derivative differences and matrix conditioning. This is a numerical diagnostic, not a rigorous error bound or a statistical rejection rule. Reconstructing the finite responses from saved quantities gives zero discrepancy; the largest local chain-rule discrepancy is 1.40×10^{-9} and the MC supply-formula discrepancy is 1.94×10^{-10} . All allocation and mixed cases have positive entries of N , and no finite fixed-price state has a flagged supplier loss.

The manuscript’s batch exhibit builder reads the saved evaluation and verification files, checks their hashes, and reconstructs the reported quantities, cost changes, weighted matrices, and block contrasts. It does not import an equilibrium engine or perform a new model solve. Table 4 reports only the three evaluated allocations; no zero-crossing location or intervening equilibrium path is inferred.

B.6 Original joint-response reconstruction and provenance

The saved joint-response exercise covers seven small examples and six HBM calibrations. It computes both buyer-shock columns in each of 13 baselines, using symmetric log-wedge steps of 10^{-4} and 5×10^{-5} , for 104 local reset solves. Both columns use the same baseline within a case. All 13 cases have accepted baselines and all local solves pass; of 45 baseline starts, 35 are accepted and ten retained as rejected. The fixed-price matrix agrees with numerical perturbations to 7.2×10^{-9} , and reset derivatives agree with implicit differentiation to 2.5×10^{-8} . In the five MC examples, the supply-side formula for N and the inverse-response difference agree to 8.7×10^{-10} . The maximum step-size difference in N is 8.3×10^{-10} . Appendix A.2 describes the independent implementations of the small bargaining examples.

The exhibit builders read saved JSON files, recompute inverse-response matrices, expenditure weights, finite output contrasts, and target deviations, and record source and table hashes. They do not call the equilibrium or calibration routines. The original engine, calibration data, and accepted result artifacts remain unchanged. The generated audit records also verify the annual sourcing arithmetic and the conditional product-volume reconciliation below. Numerical checks support the stated model calculations, while the supply data do not themselves observe H , B , or their expenditure-weighted restrictions.

B.7 Sourcing feasibility and the product-volume bridge

With heterogeneous focal shares, define

$$\lambda = \frac{m_S}{m_{\text{other}}}, \quad m_{\text{other}} = \frac{m_K S_K + m_M S_M}{S_K + S_M}.$$

For $m_{\text{other}} > 0$ and $s_S > 0$, condition (17) is equivalent to

$$\lambda \leq \frac{b_r}{1 - b_r} \frac{1 - s_S}{s_S}.$$

For annual 2025 supply ranges, 80:20 is feasible for every completion when $\lambda \leq 1.17165$, and for at least one completion up to 1.33409. The corresponding 90:10 thresholds are 0.52073 and 0.59293. Thus the common- m rejection of 90:10 can be relaxed if Samsung’s focal share is sufficiently small relative to the volume-weighted focal share at the other suppliers. These are conditional feasibility thresholds, not estimated intervals for customer shares. If $m_S = 0$, Samsung’s constraint disappears; if $m_{\text{other}} = 0 < m_S$, positive target inputs cannot be supplied.

The source product panel groups NVIDIA B300, GB300, and B300A at 4,158.72 million Gb in 2025 and AMD MI325, MI350, MI355, and MI358 at 739.584 million Gb. Their HBM3e

labels do not isolate the 12-high component or identify suppliers. The 85 audited product–period rows do not provide a measured mapping to contemporaneous bilateral deliveries. Writing the inventory and work-in-process balance as $I_q = I_{q-1} + R_q - E_q - L_q$, even known embedded use E_q would not restrict receipts R_q without bounds on initial inventory, timing, losses, and omitted uses.

If the model’s rival is assigned to the reported AMD GPU scope and common $m = 0.5$ and 80:20 are imposed, its allocated annual HBM3e 12-high shipments are 1,358–1,424.5 million Gb. The reported AMD strict-product embedded-use upper bound is 739.584, requiring a reconciliation residual of at least 618.416 million Gb. This residual can include inventory and work in process, in-transit volume, losses, omitted products, and timing differences. It is not an inventory estimate. With the absent target–Samsung link, a Samsung-specific residual of at least 367.916 is also required; the two bounds overlap and must not be added. No independently observed upper bound on these residuals is available, so this exercise does not exclude the 80:20 allocation or narrow the policy-effect range.

For completeness, Tables 16 and 17 summarize the public sourcing and contracting disclosures and the limits of the model objects they inform.

Table 16: Product-specific shared-sourcing evidence

Supplier	Accelerator family	Publicly documented object
Micron	NVIDIA B300/GB300	36GB 12-high HBM3E design-in; March 2025 announcement
Micron	AMD MI350 series	36GB 12-high HBM3E integration and compatibility work; June 2025 announcement

These positive disclosures leave bilateral quantities and the full Samsung–SK hynix–Micron adjacency unresolved. Other links are not assigned zero because an announcement is absent. The numerical buyers represent hypothetical allocations, rather than an estimated NVIDIA–AMD pair.

Table 17: Evidence and the model objects it disciplines

Evidence	Use in the analysis	Remaining unobserved object
Product design-in and qualification	Motivates restricted sourcing and a shared supplier	Complete adjacency, bilateral purchases, switching elasticity
Advance price and volume agreements	Motivates distinct contract-adjustment states	Call-off flexibility, transfers, reset frequency, bargaining weights
Product supply ranges	Common physical supply targets and supplier composition	Buyer allocations, inventories, variable costs
Broad customs records	Historical industry context	HBM bilateral prices and causal policy incidence

C Descriptive customs context

The customs data cover monthly Korean exports by destination and ten-digit HS category from January 2010 through August 2025. HS 8542323000 is a broad multi-component integrated-circuit category. It does not isolate the HBM3e 12-high product used in the calibration. Taiwan’s

share in that category and destination-level unit values provide background on the changing composition of trade, but do not recover a supplier–buyer HBM price.

Figures 2 and 3 report destination shares and unit values. Export unit values divide reported dollar value by reported kilograms; observations with zero kilograms are excluded. The plotted series use a twelve-month trailing moving average. The world total supplies the denominator for Taiwan’s share, and the “Others” unit value uses world totals minus Taiwan. Policy and technology markers are historical annotations, not instruments or an event-study identification design. Product mix, package characteristics, and destination composition can all change these statistics. The quantitative exercise does not fit them or interpret them as evidence favoring one upstream conduct rule.

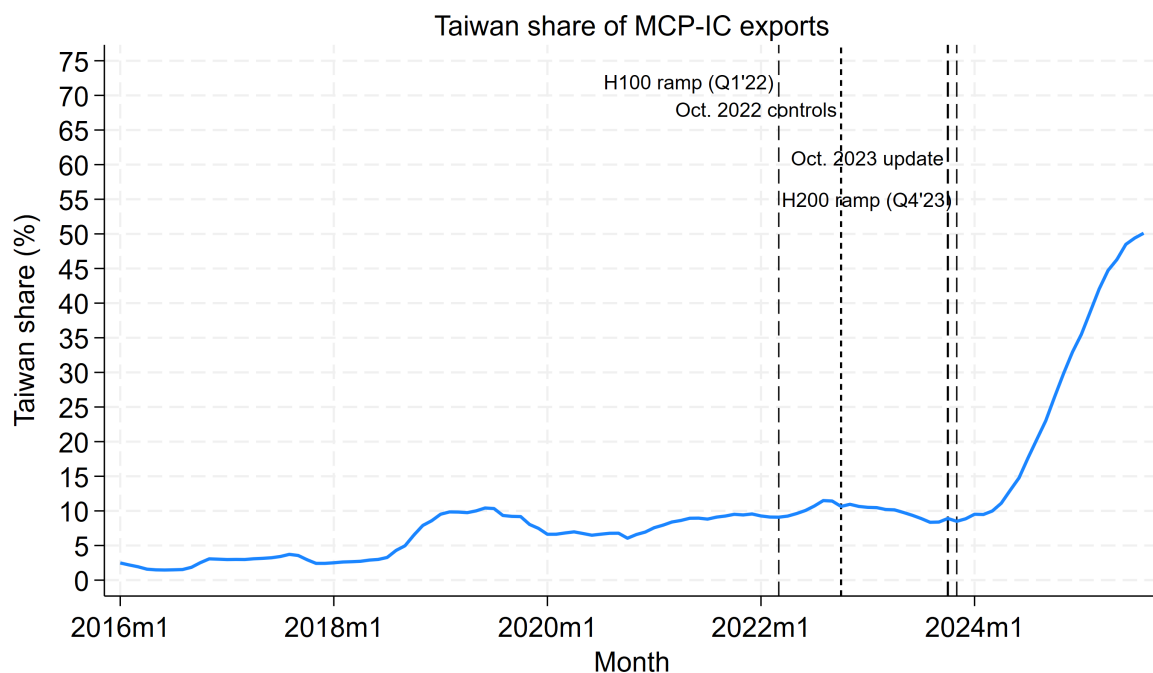


Figure 2: Taiwan share of Korean exports in HS 8542323000, a broad memory-related category.

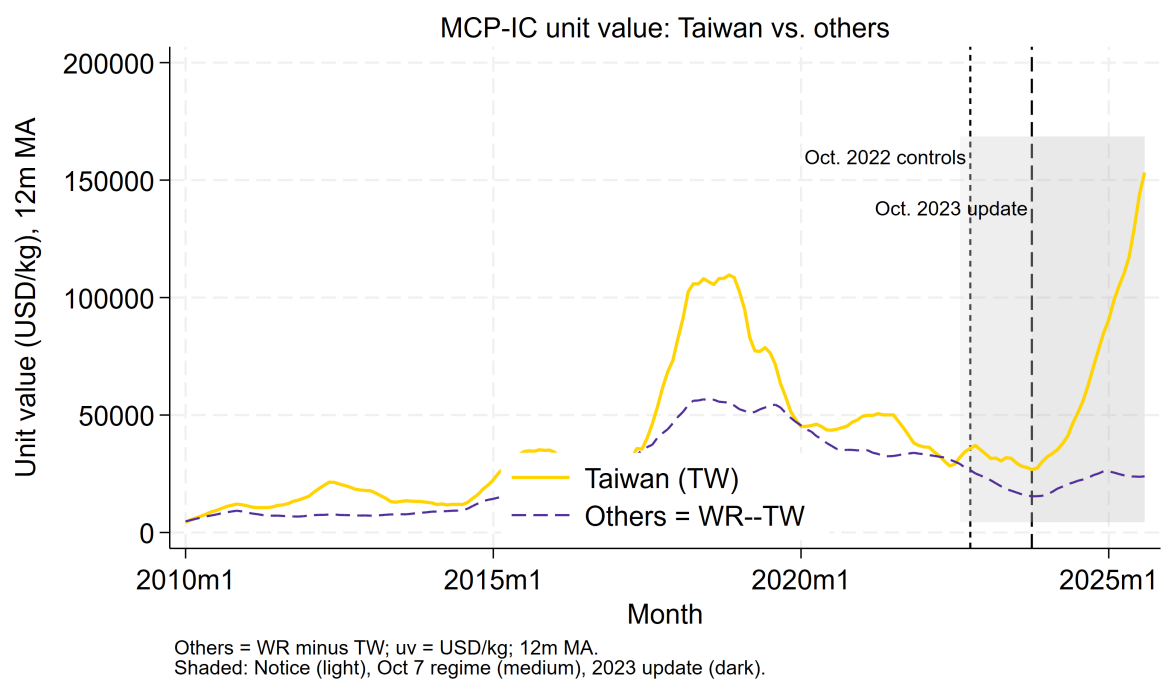


Figure 3: Destination-level export unit values in the same broad category. Dollars per kilogram do not measure a constant-quality HBM transaction price.

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